

AWS re:Invent

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ARC318

Best practices for disaster recovery of workloads on AWS

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Agenda

Disaster recovery (DR) and disaster recovery objectives

Disaster recovery strategies

Discussion and Q&A



“Everything fails, all the time.”

Werner Vogels

CTO, Amazon.com



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Disaster events

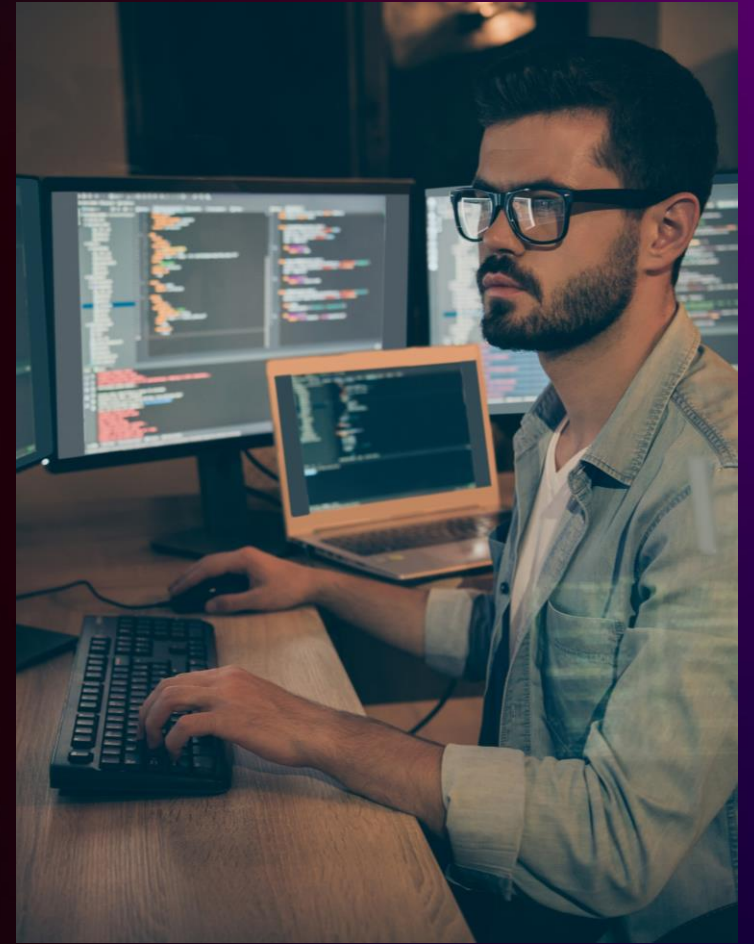
Natural disaster



Technical failure



Human actions



Understanding the business impact

Business continuity plan

How will we operate?

Risk assessment

Which risks? How likely?

Business impact analysis

What is the cost of disruption?

**Fire at data center
costs cloud provider
\$122 million**

Recovery objectives: RPO/RTO

How much data can you afford to recreate or lose?

How quickly must you recover?
What is the cost of downtime?

Recovery point objective (RPO)

Recovery time objective (RTO)



AWS Regions and Availability Zones (AZs)

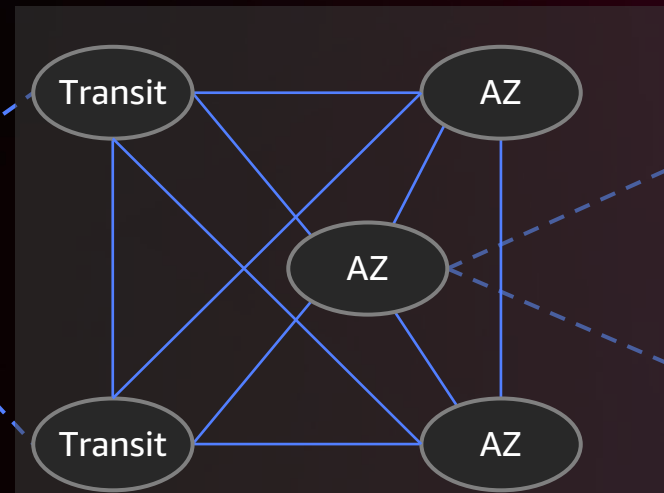
AWS REGIONS ARE PHYSICAL LOCATIONS AROUND THE WORLD WHERE WE CLUSTER DATA CENTERS

30 AWS Regions worldwide



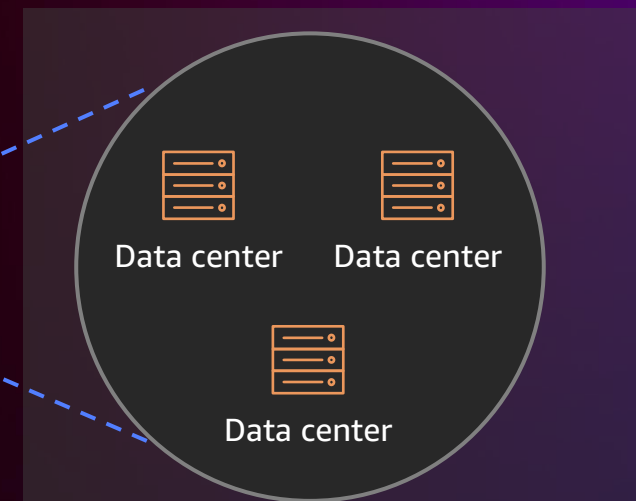
- AWS Regions
- Announced Regions

Each AWS Region has multiple AZs



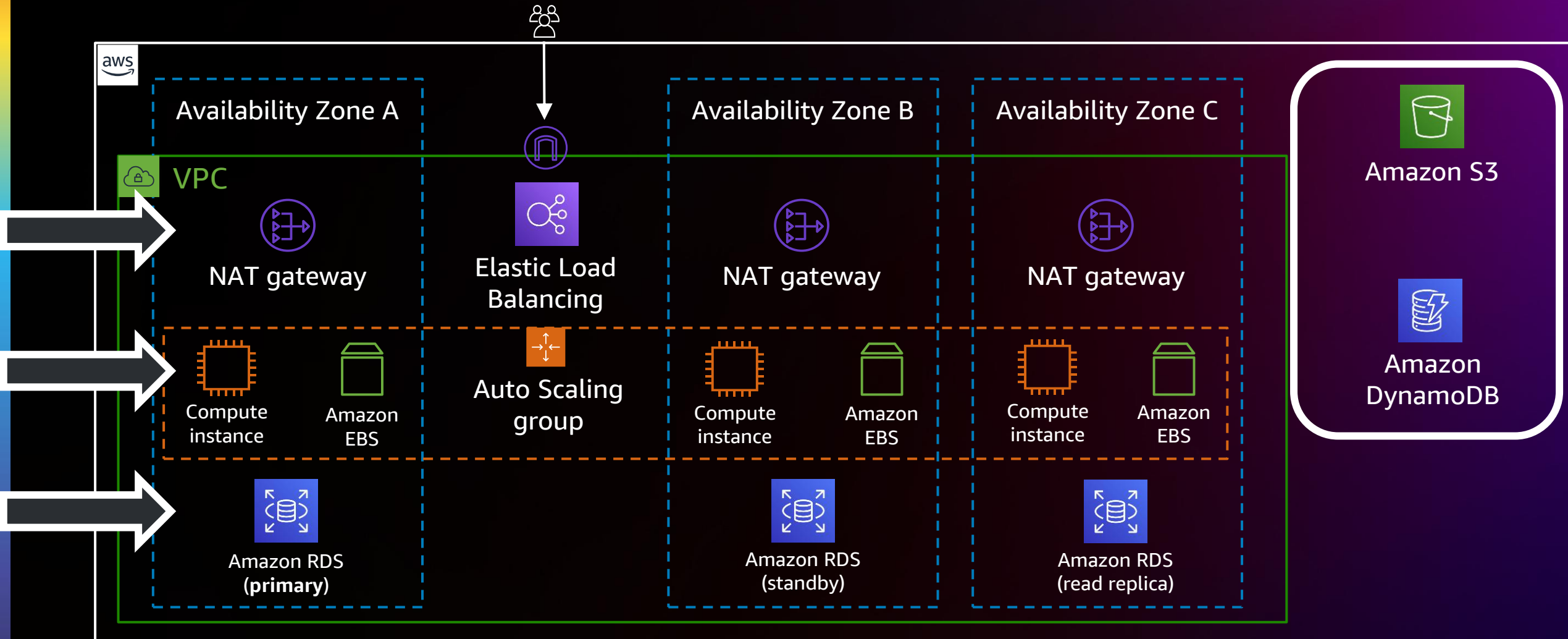
A Region is a physical location in the world

Each AZ includes one or more discrete data centers



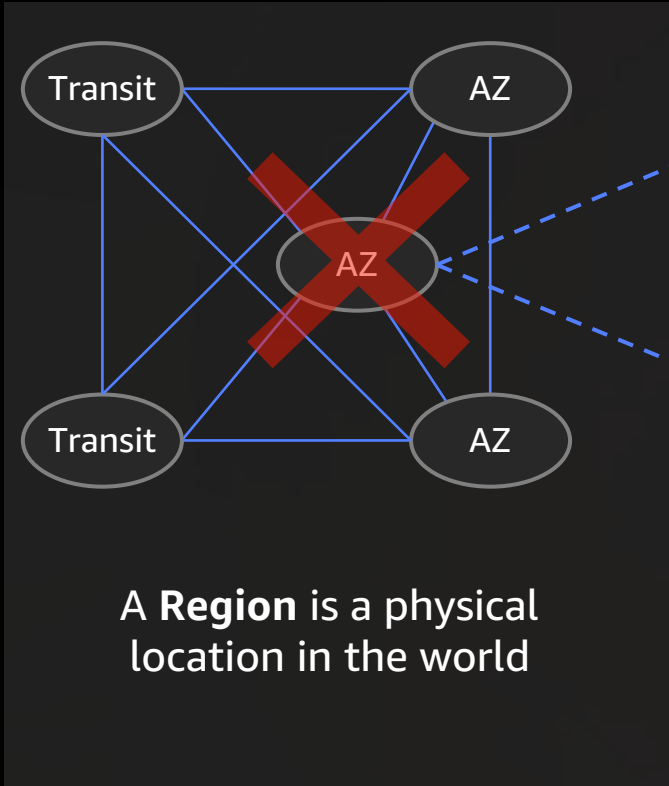
Data centers, each with redundant power, networking, and connectivity, housed in separate facilities.

Multi-AZ for high availability (HA)

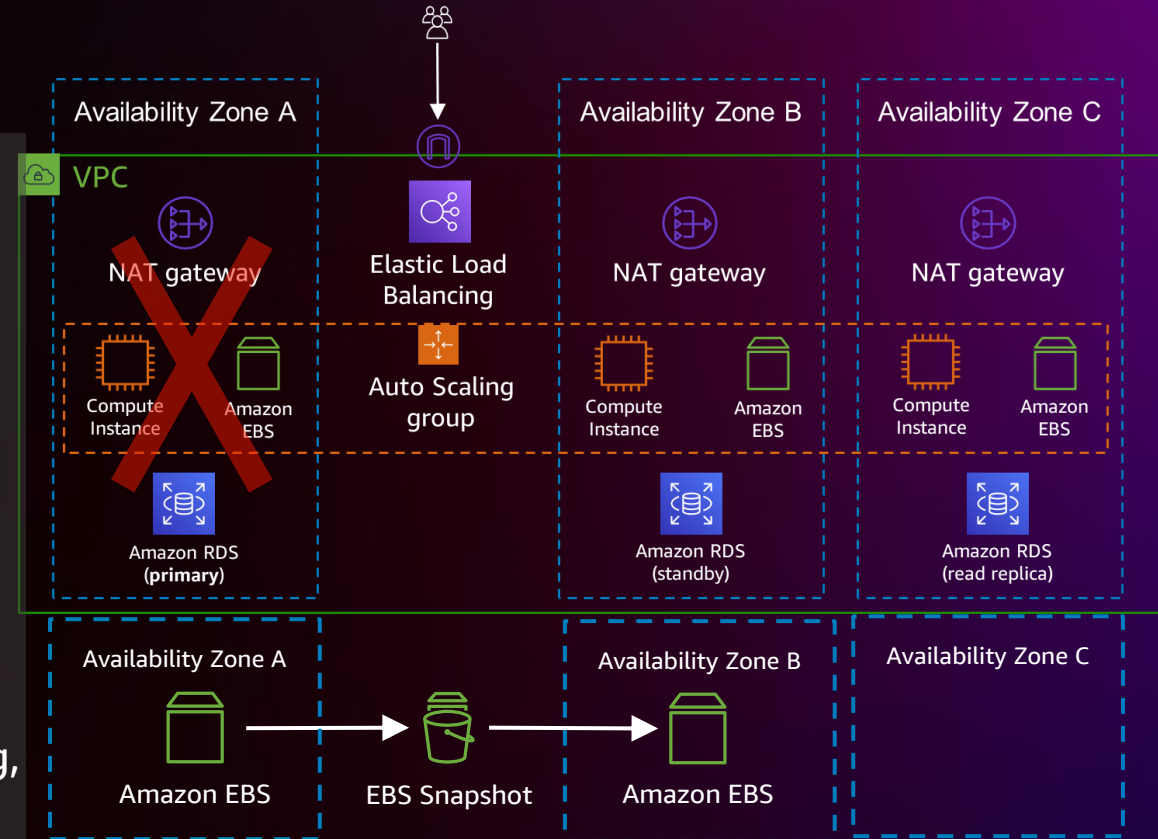
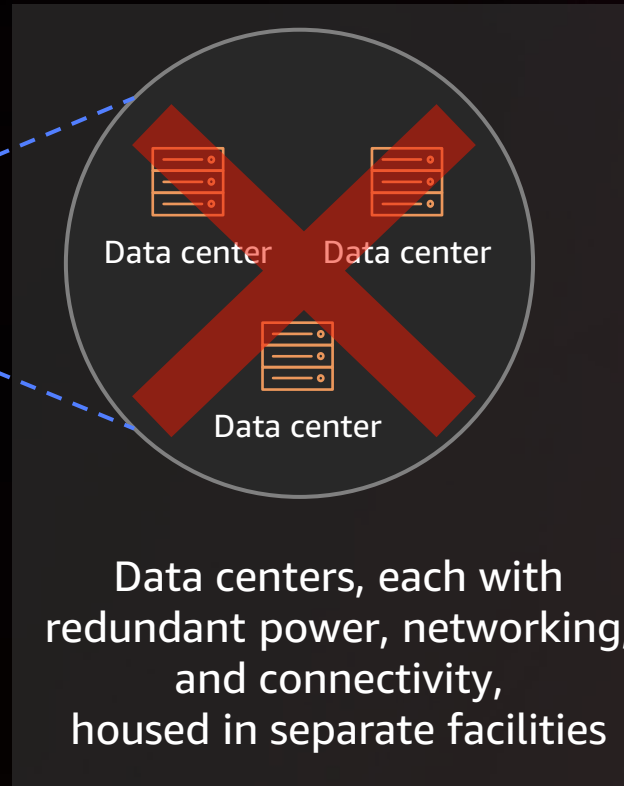


Multi-AZ for disaster recovery (DR)

Each AWS Region has multiple AZs

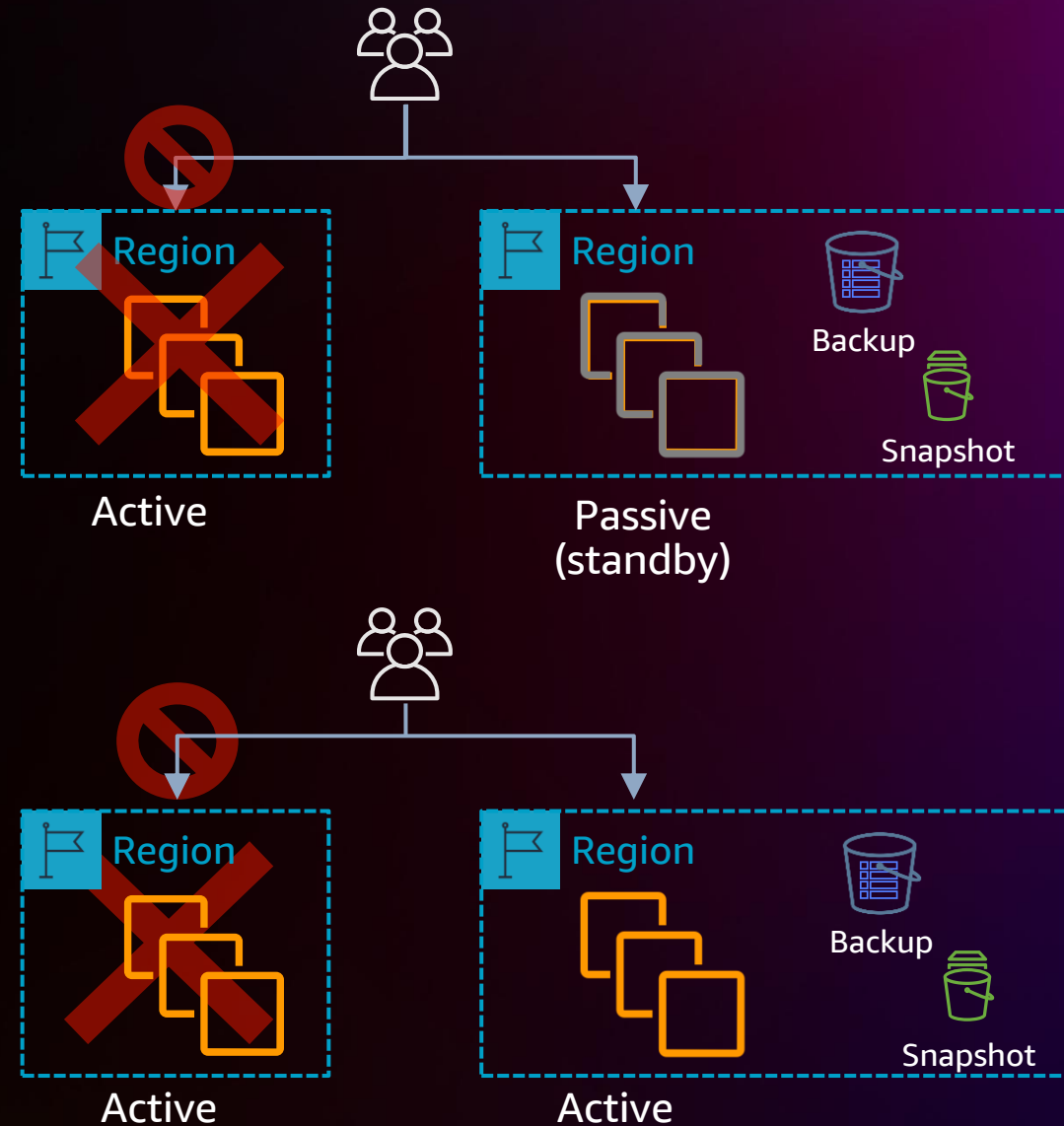
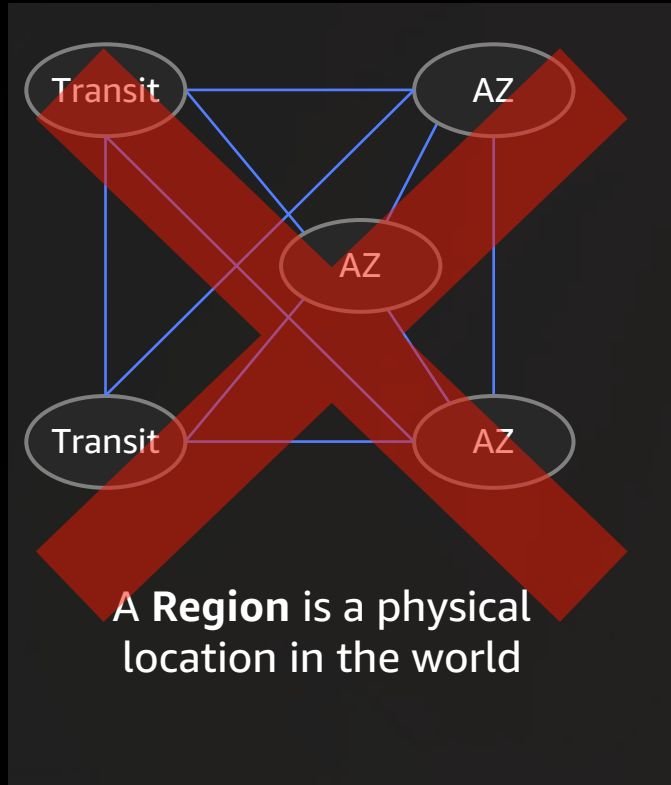


Each AZ includes one or more discrete data centers



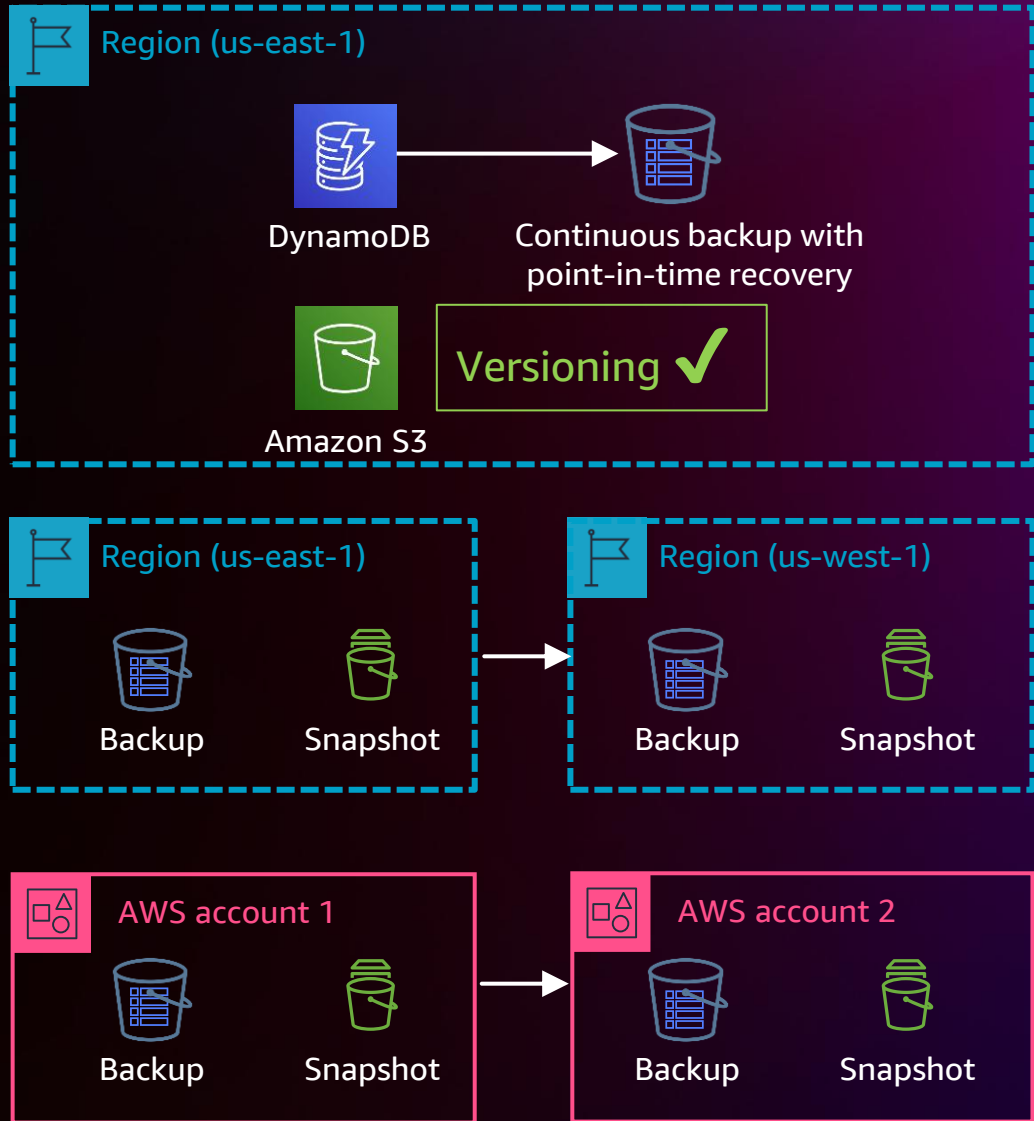
Multi-Region for disaster recovery (DR)

Each AWS Region has multiple AZs

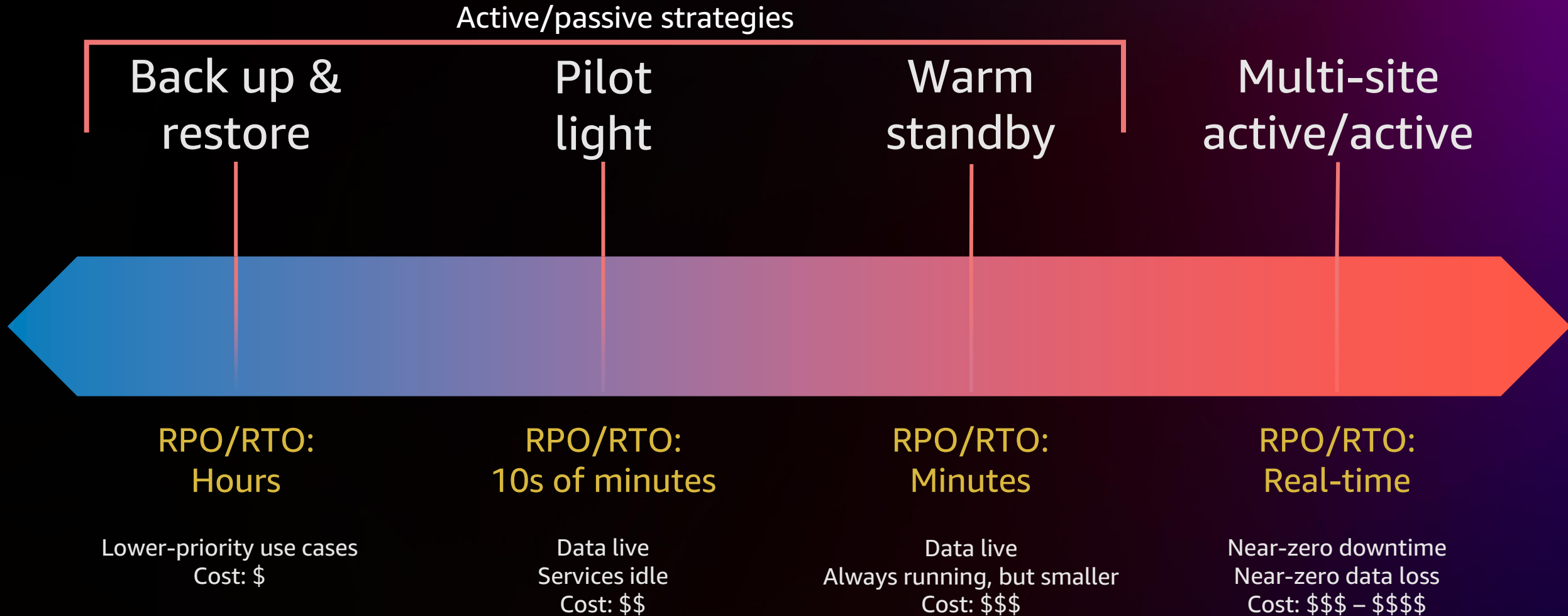


Data protection

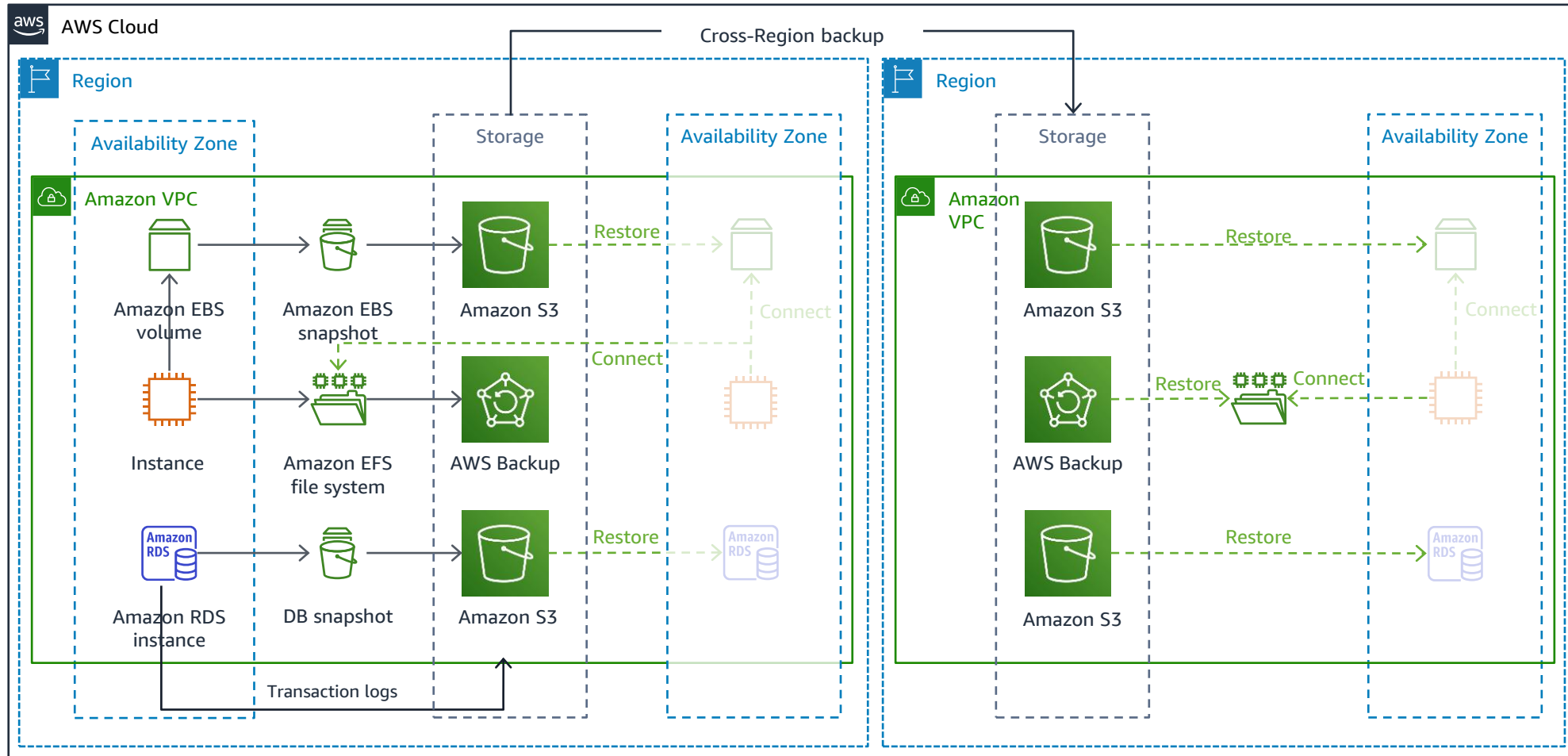
- Data corruption
- Accidental deletions
- Malicious intent



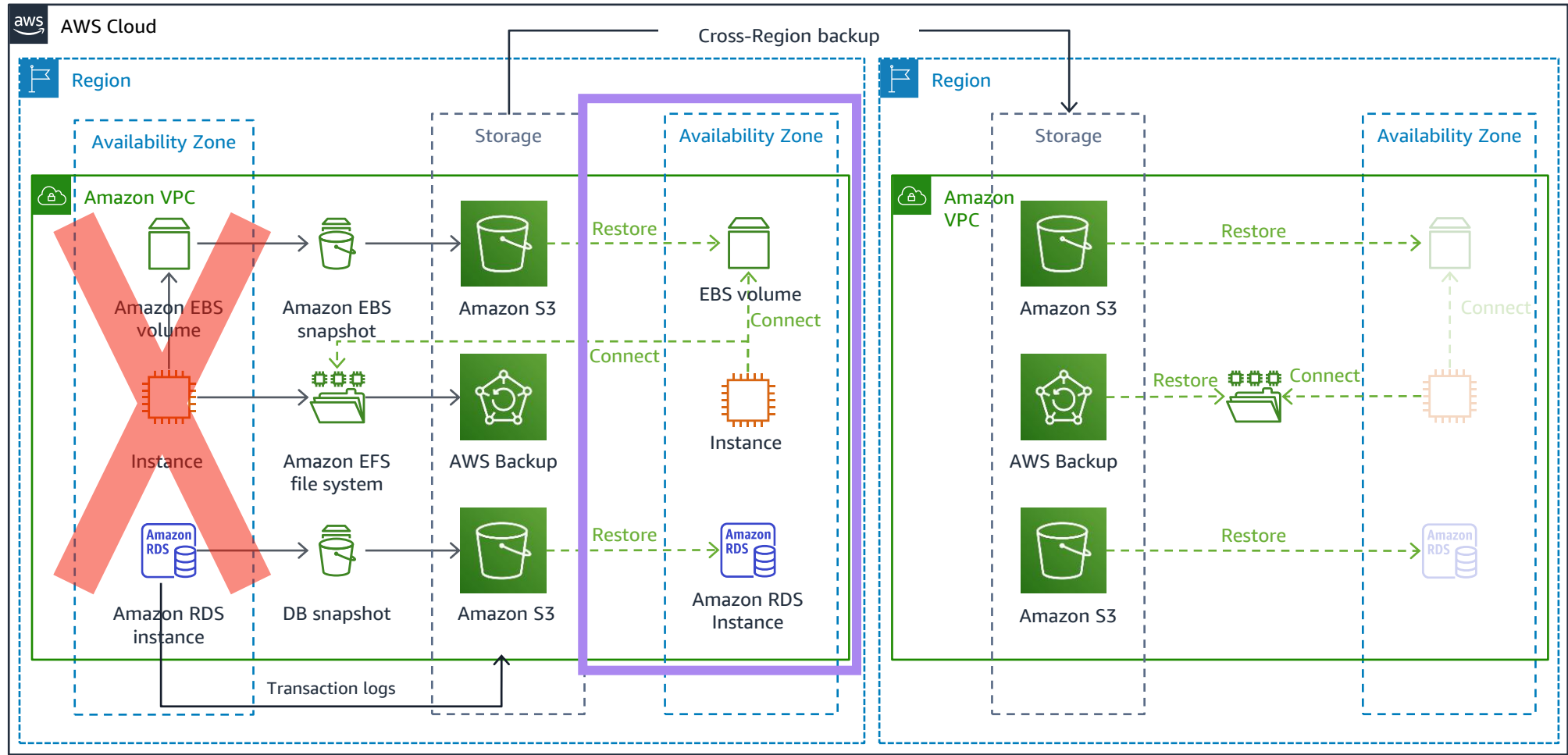
Strategies for disaster recovery



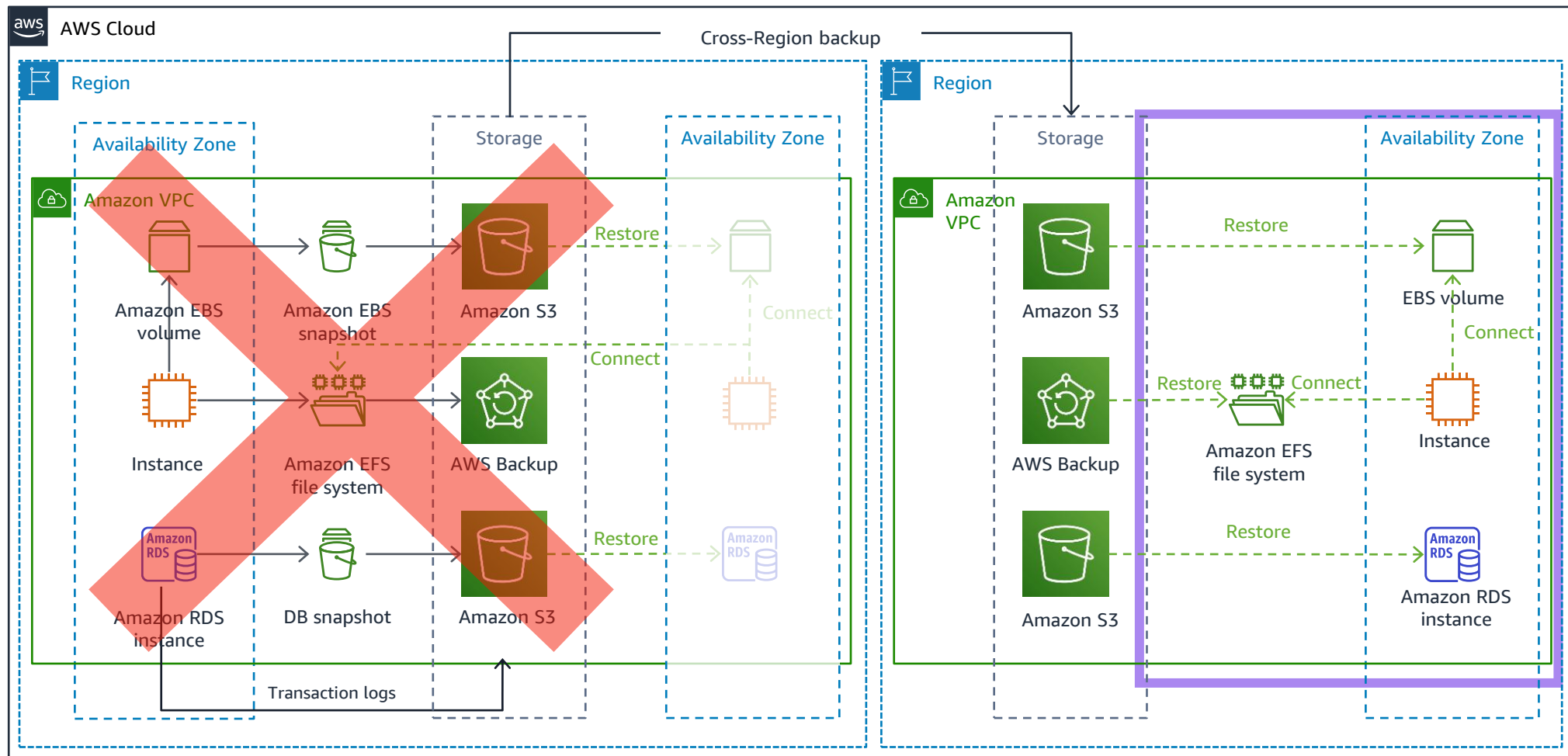
Strategy: Back up and restore



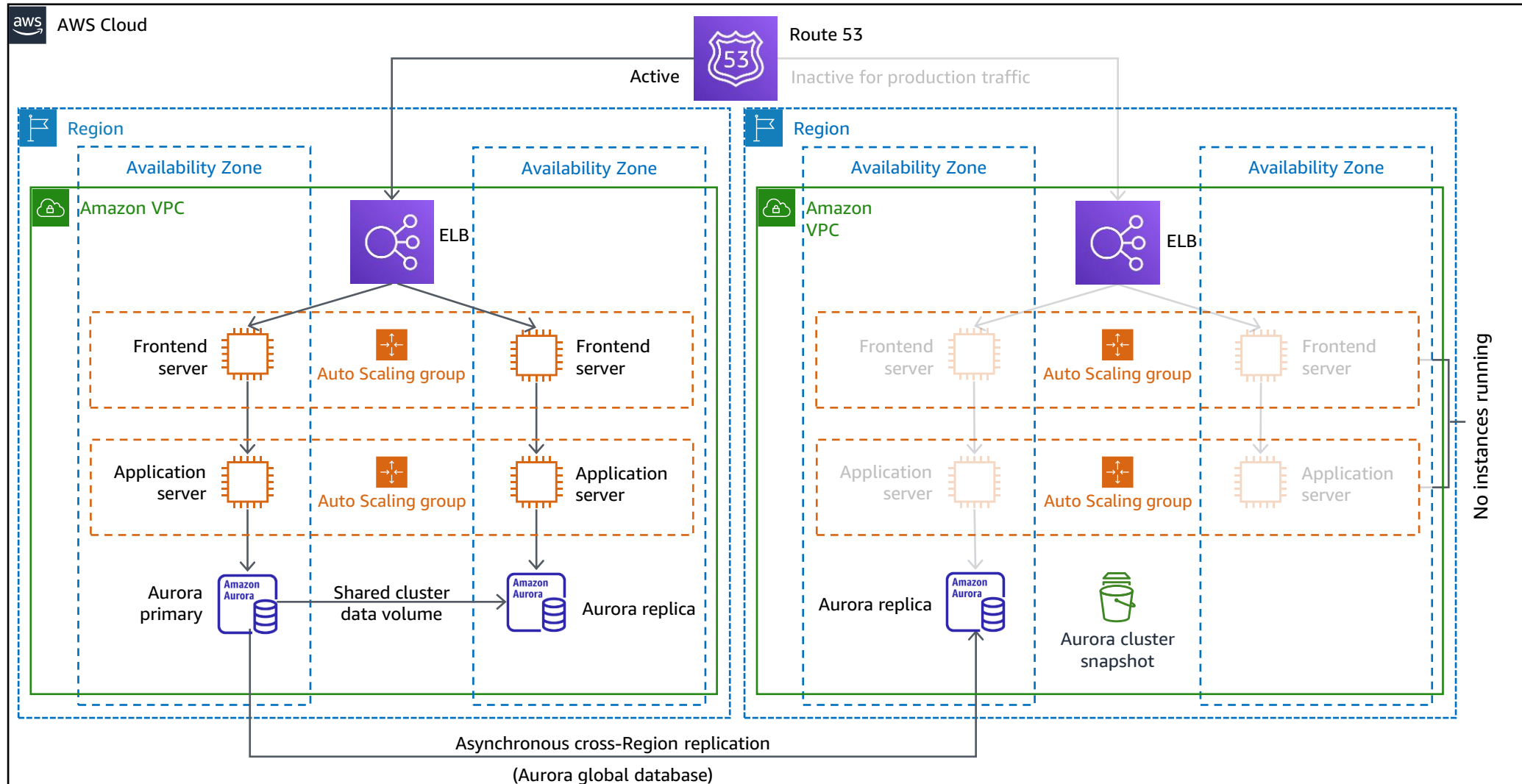
Failover: Back up and restore to AZ



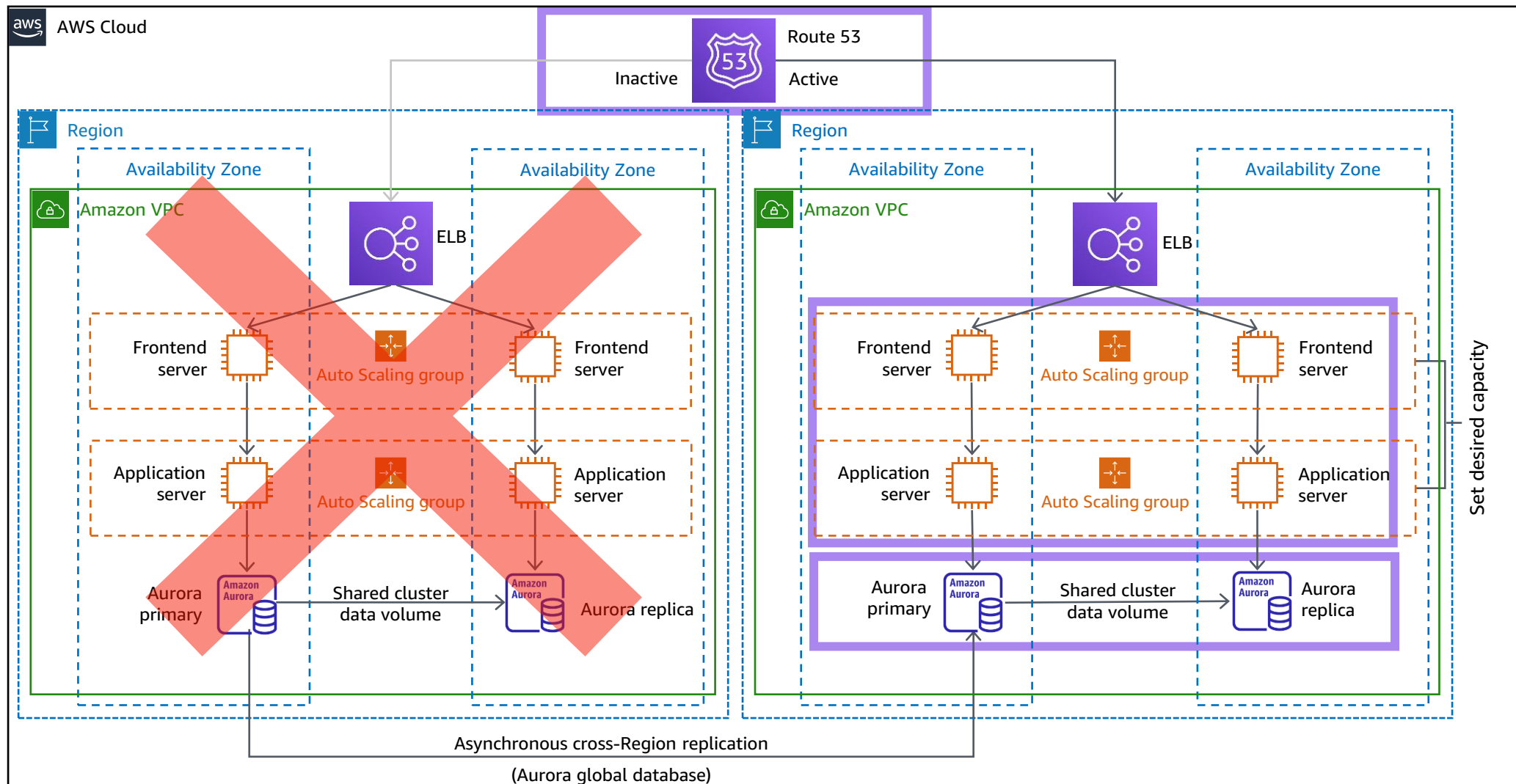
Failover: Back up and restore to Region



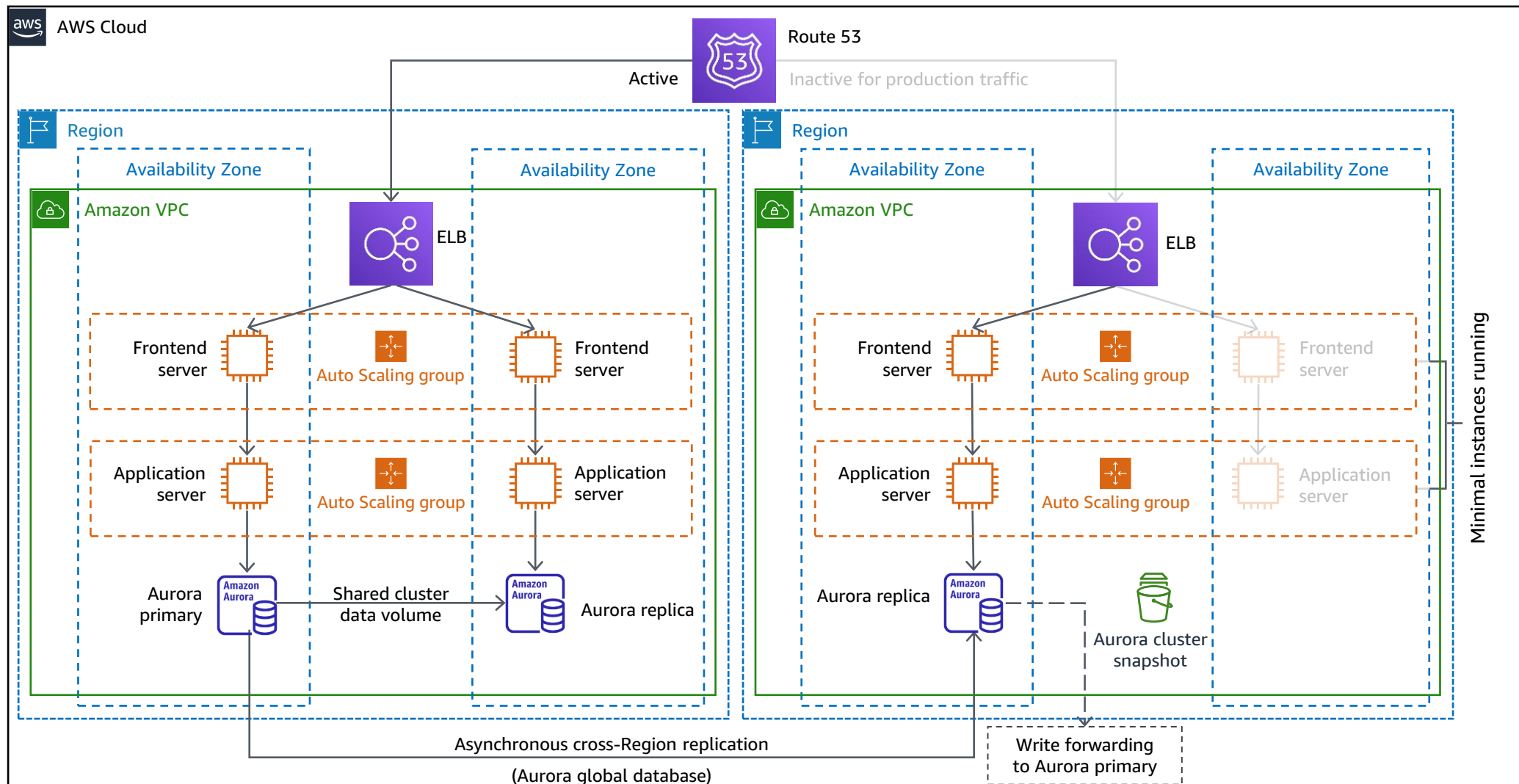
Strategy: Pilot light



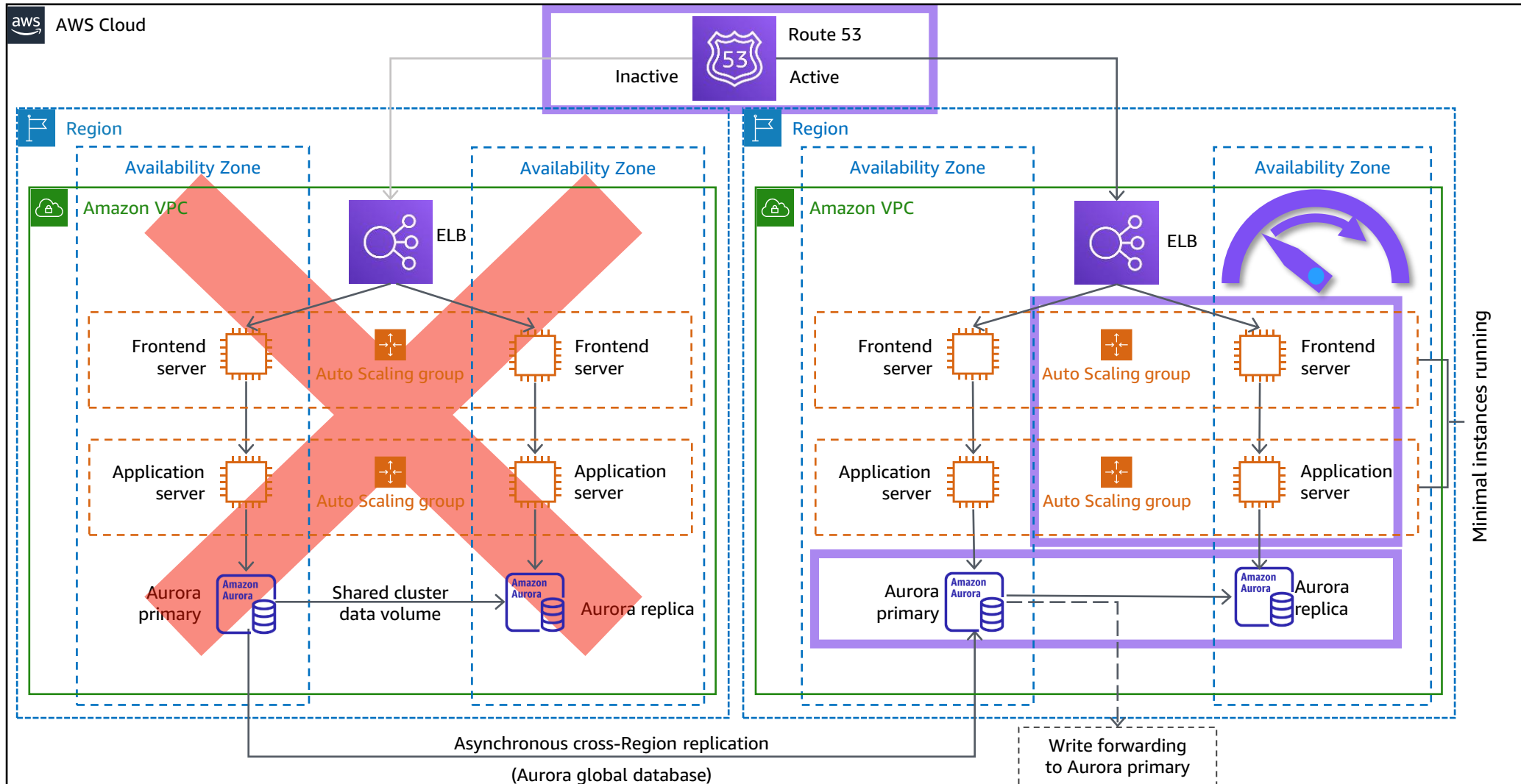
Failover: Pilot light



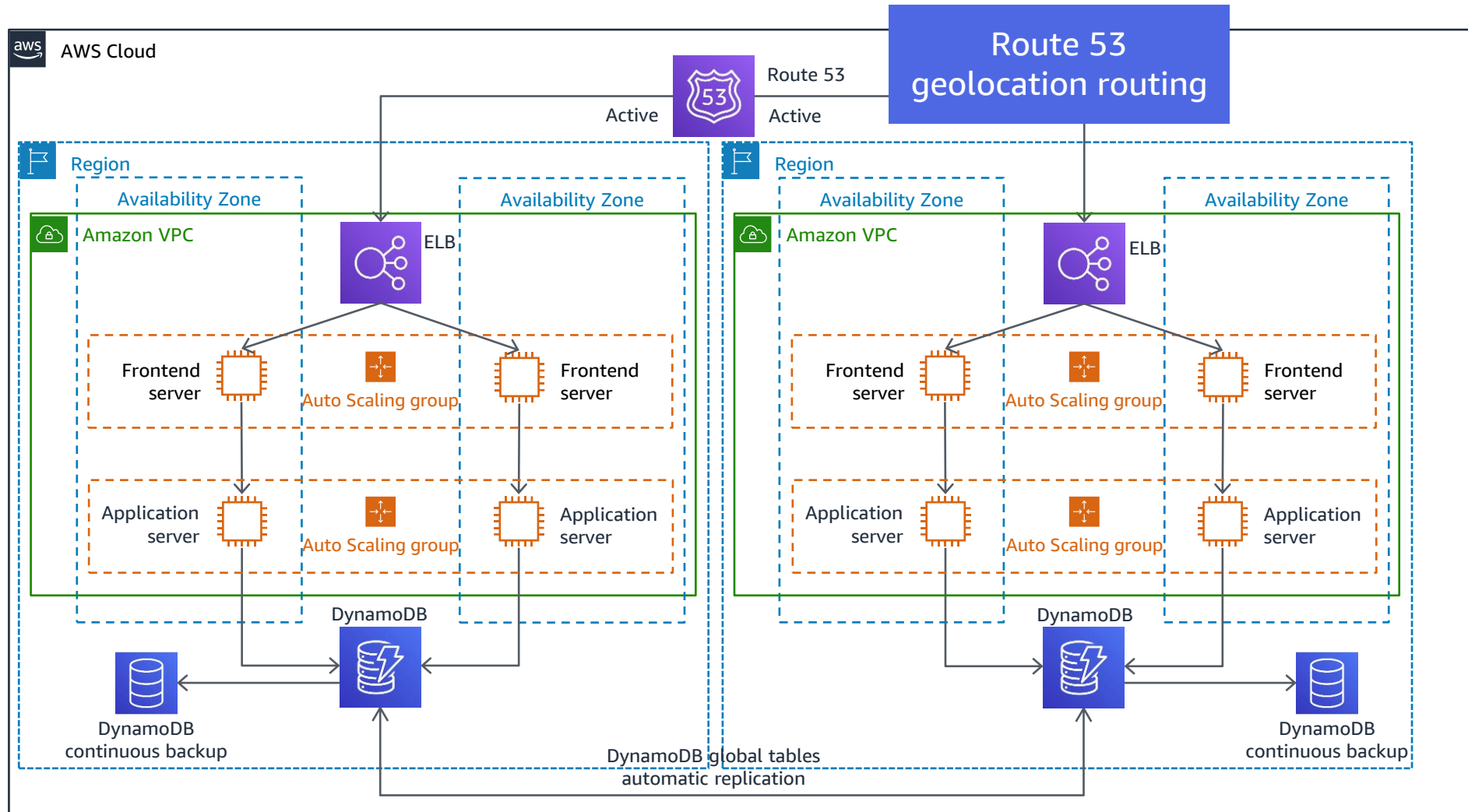
Strategy: Warm standby



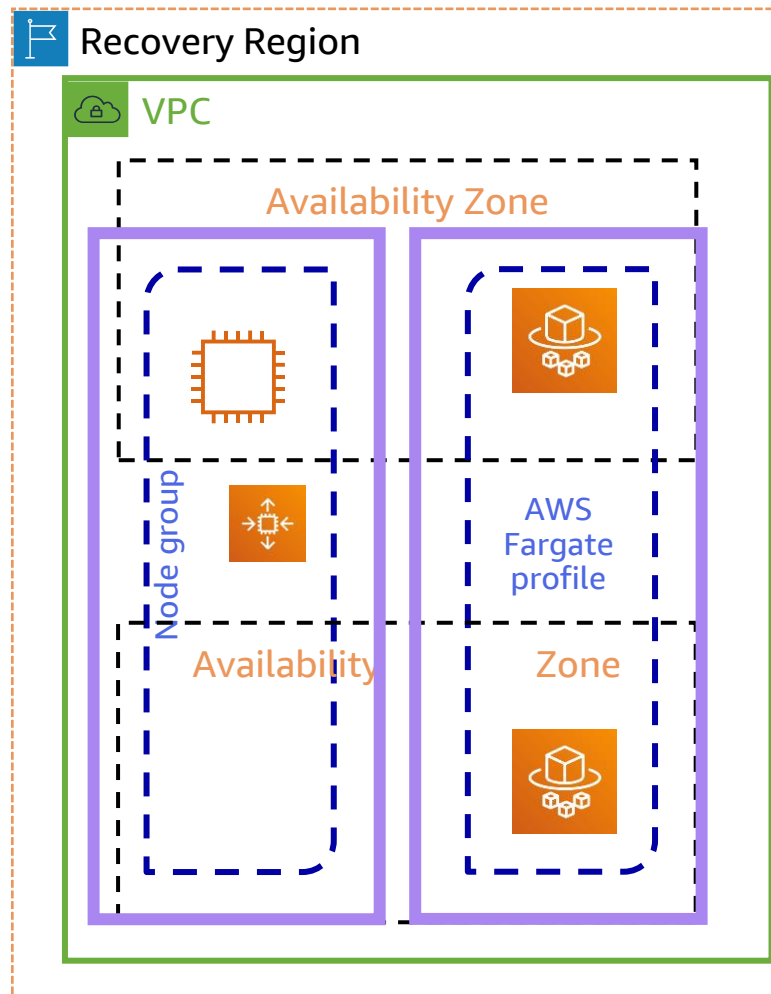
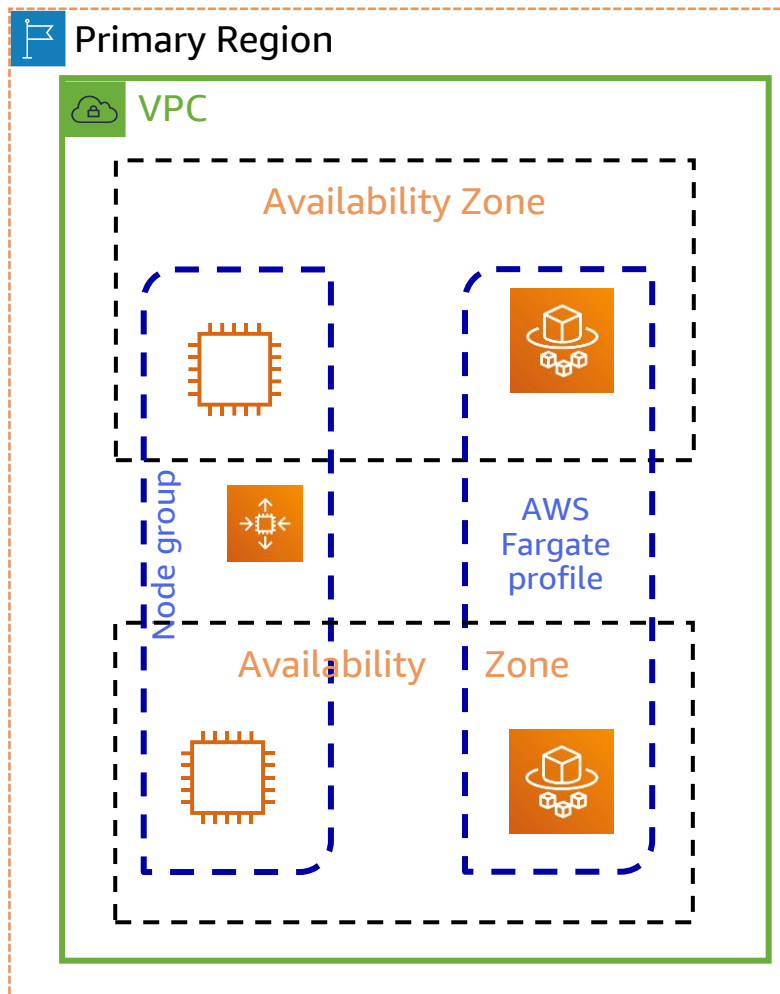
Failover: Warm standby



Strategy: Multi-site active/active



Containers: Pilot light/warm standby

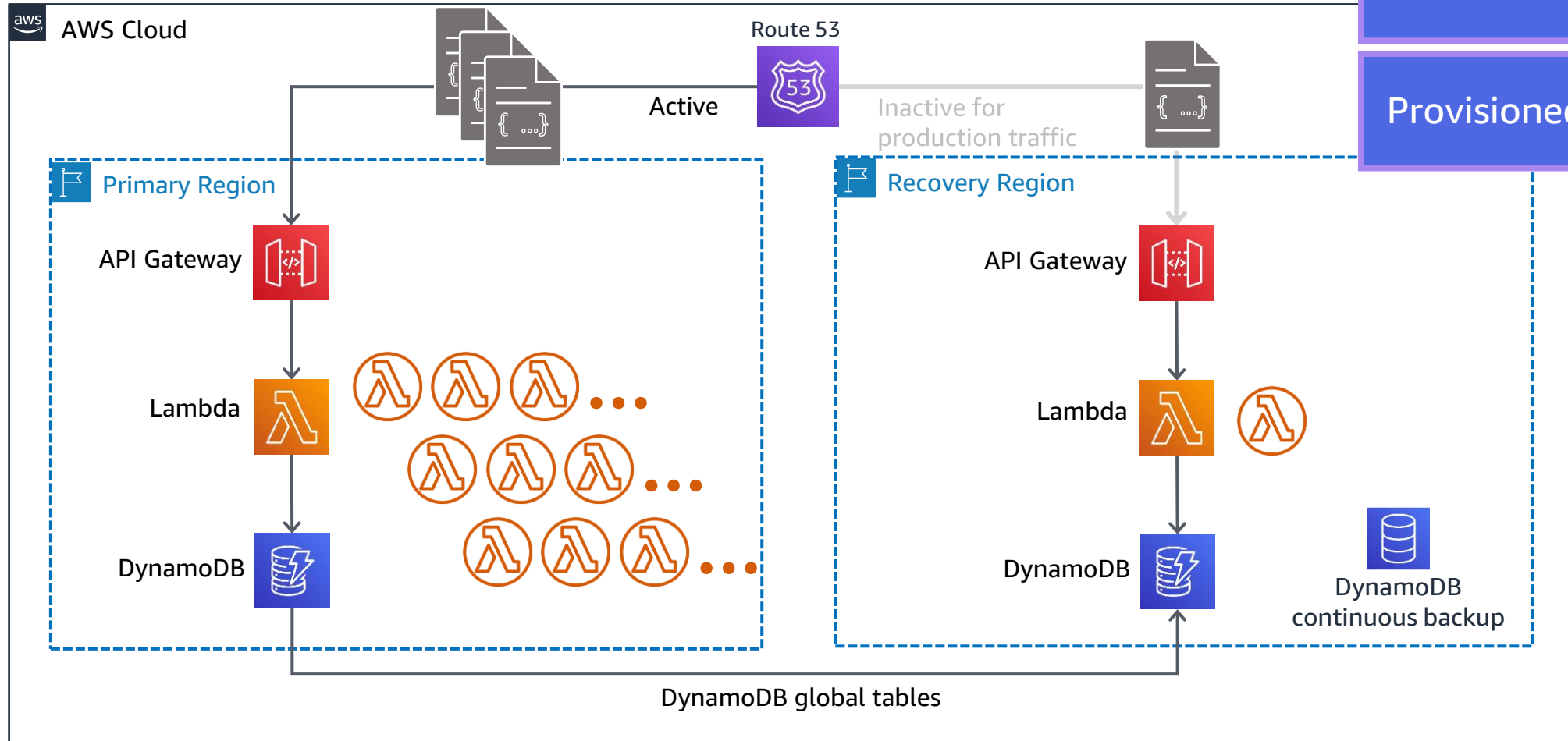


Amazon Elastic
Kubernetes Service
(Amazon EKS)

Serverless

Reserved concurrency

Provisioned concurrency



Game Days

SIMULATE FAILURE OR EVENT TO TEST SYSTEM RESILIENCY, PROCESSES, AND TEAM RESPONSES



People

Cross-disciplinary
team
Processes



Briefing

Overview
Roles



Planning

Preparation
Hypothesis



Running

Run experiment

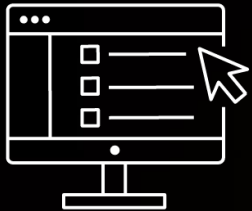


Analysis

Verify
Improve

Resilience Hub: Key capabilities

Assess



Assess whether or not
RTO/RPO is met

Can schedule daily or
run as part of CI/CD

Resilience recommendations



Optimize for...

Availability Zone RTO/RPO

Cost

Minimal changes

Operational recommendations



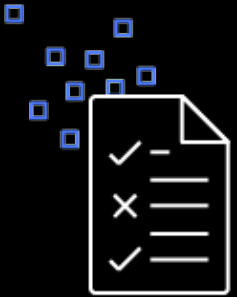
Alarms

Standard operating
procedures (SOPs)

Fault injection experiments

Route 53 Application Recovery Controller

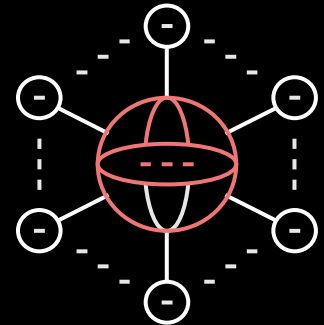
Continuously monitors an application's ability to recover from failures and controls application recovery across multiple AWS Availability Zones, AWS Regions, and on-premises environments



Readiness check



Routing control



Safety rules

Resources

ARC304 Build for resilience using Amazon Route 53 Application Recovery Controller

ARC319 Designing a multi-account environment for disaster recovery

FSI307 Advanced cross-Region disaster recovery patterns

STG312 Build a disaster recovery solution with AWS storage services

ENT223 Resiliency in the cloud: Disaster recovery with VMware Cloud on AWS



Whitepaper, "Disaster Recovery of Workloads on AWS: Recovery in the Cloud"

https://bit.ly/DR_AWS

Well-Architected Labs: Hands-on DR labs

<https://bit.ly/aws-dr-labs>

AWS Architecture blog: Disaster Recovery Series

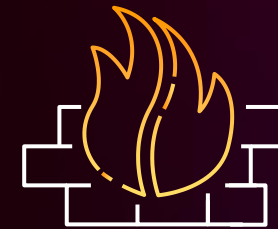
<https://bit.ly/aws-dr-blog>

AWS Well-Architected

<https://aws.com/well-architected>

AWS Solutions Library

<https://aws.com/solutions>



Disaster recovery



AWS Well-Architected

Thank you!

Seth Eliot

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Disruption types



Application

Data deletion
Data corruption
Software bugs



Infrastructure

Component failure

- Hard drive
- Power supply

In AWS, may look like

- EC2 instance failure
- RDS instance failure



Availability Zone

Natural disaster

- Fire
- Flood

Technical failure

- Power outage
- Network outage



Region

Unusual natural disaster

Technical failure:
Region-wide outage